The Probability Scale and Listing Outcomes

Name _____ Date _____ No calculator Version 1

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KEY WORDS probability *(tūponotanga)* · impossible *(tē taea)* · unlikely *(tūpono iti)* · even chance

A The probability scale ●○○

Remember: A probability is a number from 0 to 1. Impossible is 0, an even chance is $\frac{1}{2}$ and certain is 1.

1. Choose the best word for each event: impossible, unlikely, even chance, likely or certain.

a) Getting a head when a fair coin is tossed _____

b) Rolling a 6 on a fair dice _____

c) Rolling a number less than 7 on a fair dice _____

d) Rolling a number greater than 1 on a fair dice _____

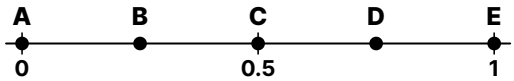
e) Rolling a 9 on a fair dice _____

f) Picking a red counter from a bag that holds 1 red and 9 blue counters _____

2. Write the probability, as a number, of an event that is:

a) impossible _____ **b)** certain _____ **c)** an even chance _____

3. Which letter on the scale shows the probability of each event?



a) Getting a tail when a fair coin is tossed _____

b) Picking a red counter from a bag that holds 1 red and 3 blue counters _____

c) Picking a blue counter from the same bag _____

d) Rolling a 7 on a fair dice _____

e) The sun rising tomorrow _____

B Listing outcomes ●●○

Remember: An outcome is one thing that can happen. Be systematic: keep the first choice fixed and go through every second choice, then move on.

4. List all the possible outcomes. Use the letters and numbers given.

a) A fair dice is rolled.

b) A coin is tossed twice (H, T).

c) A coin is tossed (H, T) and a spinner numbered 1, 2, 3 is spun.

d) You choose a sandwich (cheese C, ham H, egg E) and a drink (juice J, water W).

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5. How many outcomes are there?

a) A coin is tossed

b) A dice is rolled

c) A coin is tossed and a dice is rolled

d) 2 shirts and 3 pairs of shorts

C Outcomes in a table ●●●

Remember: A table shows every outcome of two events once. Each cell is one outcome.

6. A spinner numbered 1 to 3 and a spinner numbered 1 to 3 are spun, and the two numbers are added. Complete the table to show every possible total.

+	1	2	3
1			
2			
3			

7. Use your table.

a) How many outcomes are there altogether? _____

b) What is the largest possible total? _____

c) How many outcomes give a total of 4? _____

d) How many outcomes give a total of less than 4? _____

Extension

8. A coin is tossed three times. List all the outcomes.

9. Give an example of an event that is certain, and an example of an event that is impossible.

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The Probability Scale and Listing Outcomes

Name _____ Date _____ No calculator Version 2

Scan to practise this topic online

KEY WORDS probability *(tūponotanga)* · impossible *(tē taea)* · unlikely *(tūpono iti)* · even chance

A The probability scale ●○○

Remember: A probability is a number from 0 to 1. Impossible is 0, an even chance is $\frac{1}{2}$ and certain is 1.

1. Choose the best word for each event: impossible, unlikely, even chance, likely or certain.

a) Getting a tail when a fair coin is tossed _____

b) Rolling a 1 on a fair dice _____

c) Rolling a number greater than 0 on a fair dice _____

d) Rolling a number less than 6 on a fair dice _____

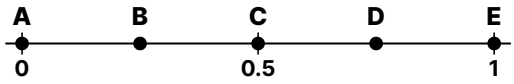
e) Rolling a 0 on a fair dice _____

f) Picking a blue counter from a bag that holds 9 red and 1 blue counter _____

2. Write the probability, as a number, of an event that is:

a) impossible _____ **b)** certain _____ **c)** an even chance _____

3. Which letter on the scale shows the probability of each event?



a) Getting a head when a fair coin is tossed _____

b) Picking a green counter from a bag that holds 3 green and 1 red counter _____

c) Picking a red counter from the same bag _____

d) Rolling an 8 on a fair dice _____

e) The day after Monday being Tuesday _____

B Listing outcomes ●●○

Remember: An outcome is one thing that can happen. Be systematic: keep the first choice fixed and go through every second choice, then move on.

4. List all the possible outcomes. Use the letters and numbers given.

a) A spinner numbered 1 to 5 is spun.

b) A coin is tossed twice (H, T).

c) A coin is tossed (H, T) and a spinner numbered 1, 2, 3, 4 is spun.

d) You choose an ice cream (vanilla V, chocolate C, mint M) and a topping (nuts N, sauce S).

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5. How many outcomes are there?

a) A coin is tossed

b) A dice is rolled

c) A coin is tossed and a dice is rolled

d) 3 shirts and 2 pairs of shorts

C Outcomes in a table ●●●

Remember: A table shows every outcome of two events once. Each cell is one outcome.

6. A spinner numbered 1 to 3 and a spinner numbered 1 to 4 are spun, and the two numbers are added. Complete the table to show every possible total.

+	1	2	3
1			
2			
3			
4			

7. Use your table.

a) How many outcomes are there altogether? _____

b) What is the largest possible total? _____

c) How many outcomes give a total of 5? _____

d) How many outcomes give a total of less than 4? _____

★ Extension

8. A coin is tossed three times. List all the outcomes.

9. Give an example of an event that is certain, and an example of an event that is impossible.

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